BANGARU SRIKARA DATTA

AI/ML Engineer

Hyderabad, Telangana, India

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and Associate Data Scientist with hands-on experience building end-to-end AI solutions, conversational chatbots, and Gen-AI/Agentic applications using Python, LangChain, Large Language Models (LLMs), and OpenAI-compatible APIs. Experienced in rapid prototyping, workflow automation, and deploying interactive AI-powered dashboards via Streamlit. Skilled in NLP, deep learning, EDA, and data visualization . Adept at cross-functional collaboration, translating business requirements into AI-driven solutions, and documenting scalable AI pipelines. Eager to accelerate AI adoption across customer success, support, and implementation workflows.

TECHNICAL SKILLS

AI / Gen-AI / Agentic Tools: LangChain, Large Language Models (LLMs), Diffusion Models, Multimodal AI, OpenAI API (compatible), Prompt Engineering

Machine Learning: Scikit-learn — Regression, Classification (Decision Tree, KNN, Random Forest, Gradient Boosting), Clustering, Ensemble Methods

Deep Learning & NLP: ANN, NLP Techniques, TensorFlow (familiar), Generative AI Concepts

Programming & APIs: Python (OOP, File Handling, Exception Handling), SQL (MySQL), FAST APIs, Streamlit

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, EDA, Statistical Modeling, Feature Engineering

Prototyping & Deployment: Streamlit (web app deployment), Optuna (Hyperparameter Optimization).

Computer Vision: OpenCV, MediaPipe — Real-time gesture recognition & multimodal AI applications

Database: MySQL — DDL, DML, CTEs, Joins, Views, Stored Procedures, Triggers, Relational Database Design

PROJECTS & AI PROTOTYPES

AI-Powered Chatbot & Delivery Time Prediction (End-to-End ML App)

Technologies: Python, Scikit-learn, Decision Tree Regressor,Manhatten Formula, LangChain, Streamlit, Pandas

- Designed and deployed a production-ready AI chatbot and ML pipeline as an interactive web application using Streamlit, simulating real-world customer-facing AI solutions similar to support/implementation chatbot use cases.
- Built an end-to-end regression model to predict food delivery ETAs for Swiggy, improving customer satisfaction through reliable AI-driven insights — directly applicable to post-sales customer success workflows.
- Performed EDA and feature engineering on structured datasets; applied Label Encoding for categorical variables (traffic, weather) and computed real-time delivery distance using the Manhatten formula from raw GPS coordinates.
- Optimized Decision Tree Regressor to model non-linear relationships; achieved $R^2 = 0.77$ on test set with low MAE, demonstrating model evaluation and iteration aligned with rapid prototyping best practices.
- Documented the full ML pipeline with clear code structure and deployment guide, demonstrating knowledge transfer and documentation skills critical for scaling AI pilots.

Gesture-Controlled Virtual Calculator — Multimodal AI & Computer Vision

Technologies: Python, OpenCV, MediaPipe, NumPy, Optuna, Real-Time AI

- Developed a contactless, AI-powered multimodal application enabling users to perform math operations via real-time hand gesture recognition — showcasing agentic AI and human-computer interaction capabilities.
- Integrated MediaPipe to detect and track 21 hand landmarks for precise fingertip-based input recognition; implemented virtual click detection using landmark distance measurements, demonstrating deep AI/ML prototyping skills.
- Designed an interactive GUI overlay on live camera feed using OpenCV for real-time user feedback — illustrating ability to build fast, working AI demos and proof-of-concept prototypes.

- Achieved 99.91% gesture-to-text accuracy; validated and optimized model parameters using Optuna's hyperparameter optimization framework, with performance tracked and documented for iteration.
- Applicable to automation workflow design: gesture inputs as a proxy for agentic triggers/workflow automation concepts.

Grocery Store Management — Relational Database & Business Analytics

Technologies: MySQL, ER Modeling, DDL/DML, CTEs, Joins, Views, Stored Procedures, Power BI

- Designed and implemented a normalized 6-table relational database to support inventory, sales, and supplier management — demonstrating ability to work with structured data systems supporting enterprise SaaS workflows.
- Performed statistical data analysis using SQL CTEs for time-series trends (monthly, weekday vs. weekend) and ranked top-performing employees and suppliers, generating actionable business intelligence dashboards in Power BI.
- Generated business metrics such as average order value per employee to support management decisions; enforced full data integrity with PK/FK constraints — aligned with customer success and operational reporting use cases.
- Created documentation and process maps for database schema and query logic, demonstrating knowledge transfer skills for scaling data solutions across teams.

EDUCATION

Bachelor of Technology — Petroleum Engineering *2020 – 2024*

Vignan's University for Science and Technology, Guntur | CGPA: 8.0 / 10.0

Relevant Coursework: Data Structures, Statistics, Python Programming, Machine Learning, AI Concepts, Data Analysis with MySQL, Data Visualization

CERTIFICATIONS

- Python Programming Certification
- Machine Learning Certification
- Data Analysis with MySQL
- Data Analysis with Power BI

KEY STRENGTHS & ATS KEYWORDS

Generative AI (Gen-AI) • Agentic AI Tools • LangChain • LLMs • OpenAI API • Rapid Prototyping • Chatbot Development • Workflow Automation • AI Dashboards • Python • FAST APIs• NLP • Deep Learning • Process Discovery • Pilot Deployment • Prompt Engineering • Prompt Libraries • Knowledge Sharing • Cross-Functional Collaboration • EDA • Statistical Modeling • SQL • Streamlit • End-to-End ML Pipeline